Smart Chilli Crop Intelligence System Using an Autonomous Ground Rover

R26-IT-075

Our Team



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Introduction to the overall project

- The project develops a Smart Chilli Crop Intelligence System using an autonomous ground rover.
- Chilli cultivation requires accurate monitoring of soil, moisture, disease, stress and growth conditions.
- The rover collects field data through cameras, GNSS/IMU sensors and soil sensors.
- AI and machine learning models analyse crop health, soil status, irrigation needs and field zones.
- The final dashboard provides farmers and agronomists with clear, practical crop management recommendations.

Research Problem

- Chilli cultivation in Sri Lanka is highly affected by disease, water stress, soil imbalance and uneven field conditions.
- Farmers still depend mainly on manual observation, fixed irrigation schedules and delayed soil testing.
- Early disease symptoms, nutrient deficiencies and moisture stress are often missed until crop damage has already occurred.
- Ground rover navigation can also become unreliable when GPS/GNSS signals degrade under crop canopy, trees or field obstructions.
- Existing solutions usually handle disease detection, irrigation, soil analysis or mapping separately, rather than as one integrated system.

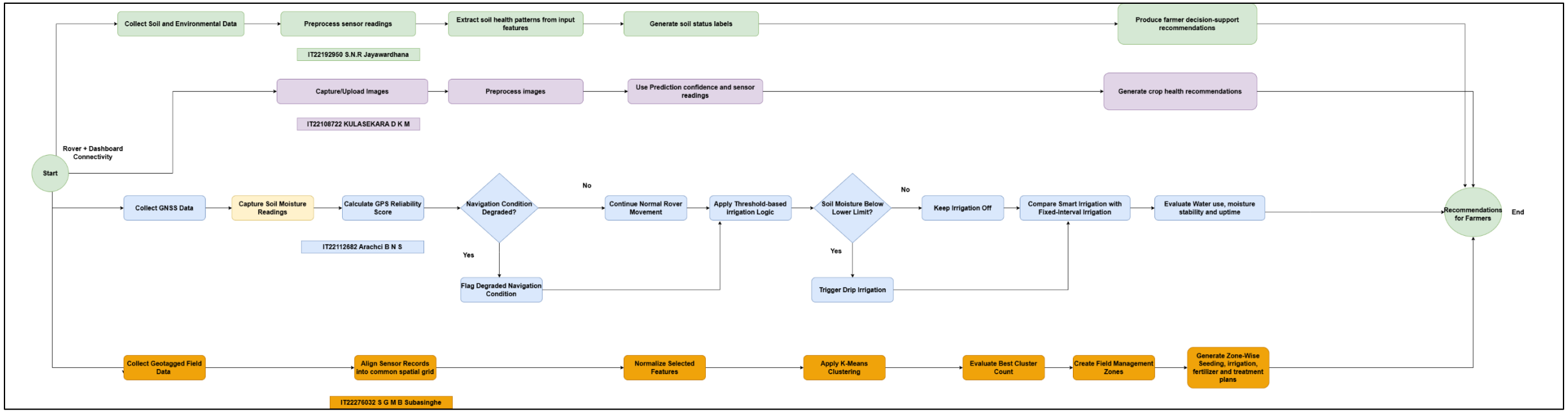
Research Question

How can an autonomous ground rover-based intelligent system improve chilli crop monitoring and decision-making by integrating disease detection, soil health prediction, irrigation control, and field zone mapping?

Research Objectives

- To design and develop a Smart Chilli Crop Intelligence System using an autonomous ground rover.
- To monitor rover navigation reliability and support sensor-based drip irrigation decisions.
- To detect chilli crop diseases, stress conditions and growth stages using AI-based image analysis.
- To predict soil health conditions using real-time soil sensor readings and machine learning models.
- To divide the chilli field into meaningful management zones using spatial clustering techniques.
- To generate zone-specific recommendations for irrigation, fertilizer use, seeding density and disease treatment.
- To present all results through a farmer-friendly dashboard for better decision-making.

Complete Flow Diagram



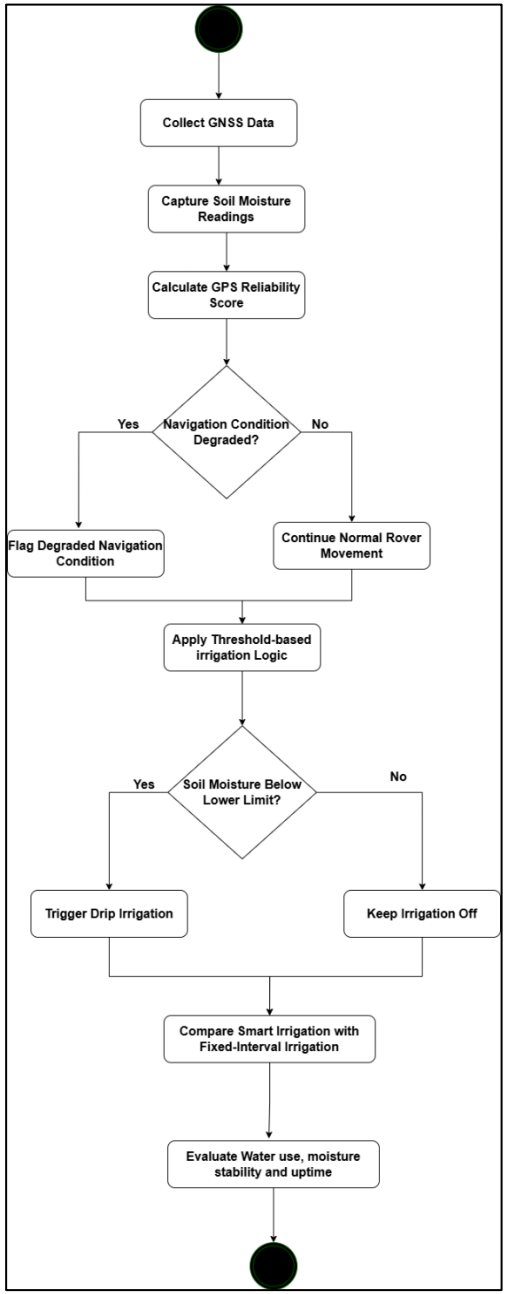


Intelligent Navigation and Sensor-Based Irrigation Control

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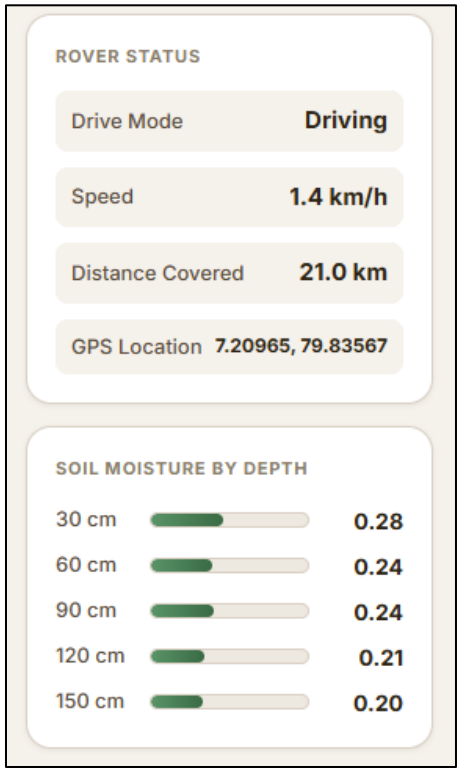
Specialization - Information Technology

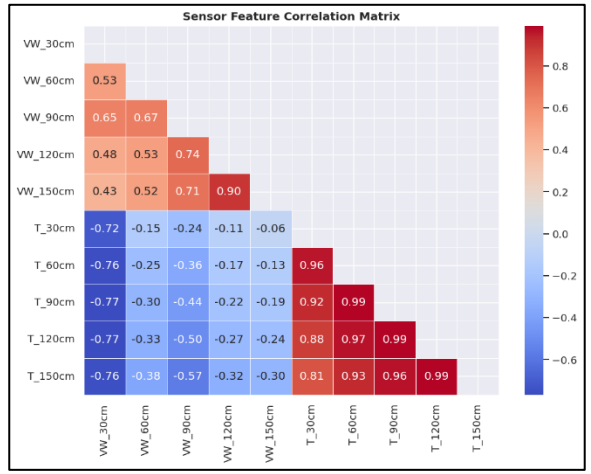
Methodology



Completion of the project

- Soil moisture anomaly detection model pipeline was rebuilt and tested using field-scale sensor data.3,346 sensor records were cleaned and transformed into 35 engineered features.
- Isolation Forest was used to identify abnormal moisture patterns, and Random Forest was trained for anomaly prediction.
- Model files, scaler, feature list and an interactive demo GUI were prepared for live demonstration.
- The output now supports rover-based moisture monitoring, irrigation alerts and dashboard integration.







AI-Based Chilli Crop Disease, Stress and Growth Monitoring

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Research Question, Specific and Sub Objectives

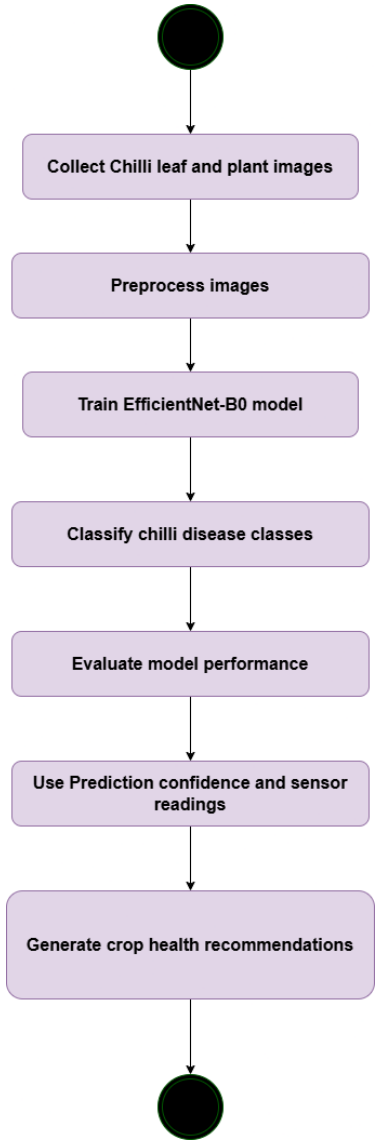
Sub-Research Question

How can an AI-based ground rover system detect chilli crop diseases, stress conditions and growth stages using image analysis and sensor data?

Sub-Objectives

- ❑ Develop an image classification model to identify major chilli crop diseases.
- ❑ Detect plant stress conditions using visual symptoms and environmental data.
- ❑ Classify chilli plant growth stages from rover-captured field images.
- ❑ Apply Grad-CAM explainability to show disease-affected image regions.
- ❑ Generate crop health outputs for field zone mapping and soil intelligence components.

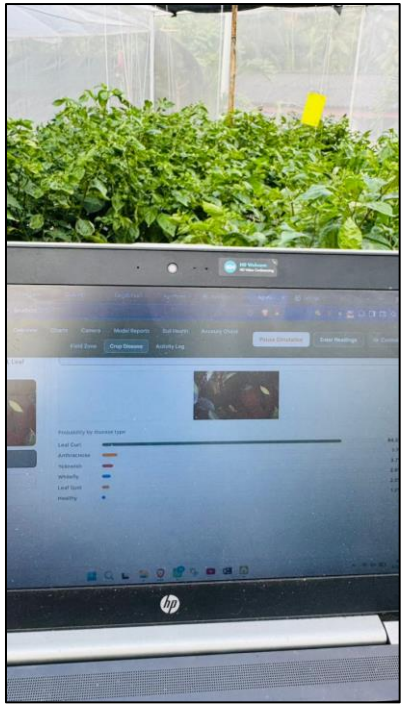
Methodology



Completion of the project

- Chilli disease detection model pipeline was completed using an 8-class chilli plant disease dataset.
- Dataset preprocessing, class mapping, train/validation/test splitting and image augmentation were implemented.
- EfficientNet-B0 CNN model was developed with a custom classification head for disease prediction.
- Model evaluation outputs were generated to identify class-wise performance and improvement areas.
- Model export, class label saving, TFLite conversion and interactive disease prediction demo were prepared.





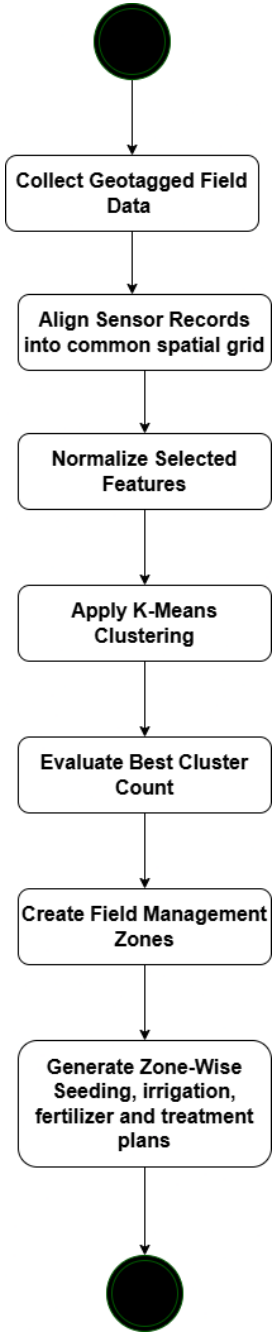


Field Zone Mapping and Autonomous Seeding and Treatment Planning

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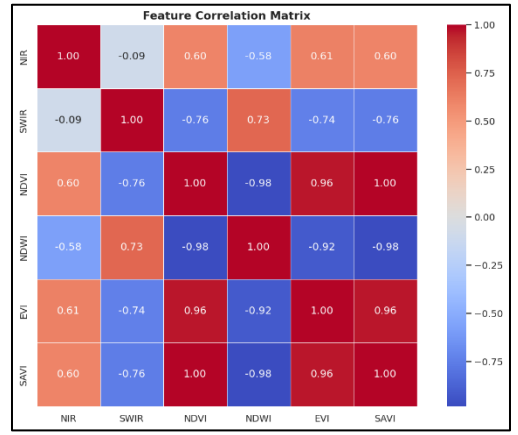
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Methodology

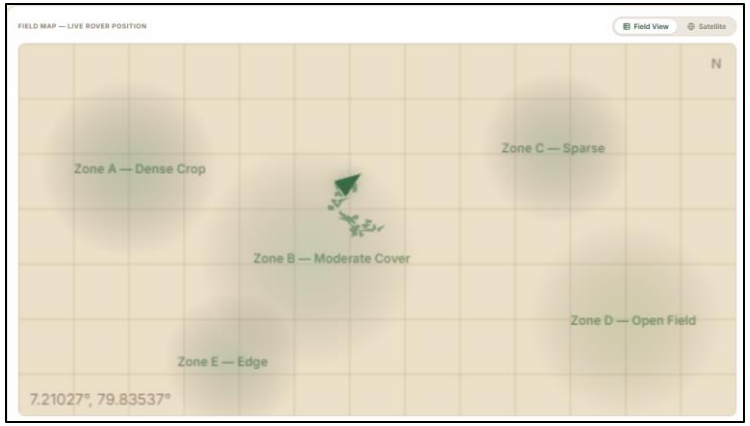


Completion of the project

- Field zone mapping ML Pipeline was completed using Sentinel-2 multispectral crop and land-cover data.120,130 cleaned records were processed using NIR, SWIR, NDVI, NDWI, EVI and SAVI features.
- K-Means clustering was applied, and optimal cluster selection was evaluated using elbow and silhouette analysis.
- Final zone segmentation achieved a silhouette score of 0.5423 with two major field-zone groups.
- Zone distribution, centroid profiling and interactive field-zone mapping demo were prepared.







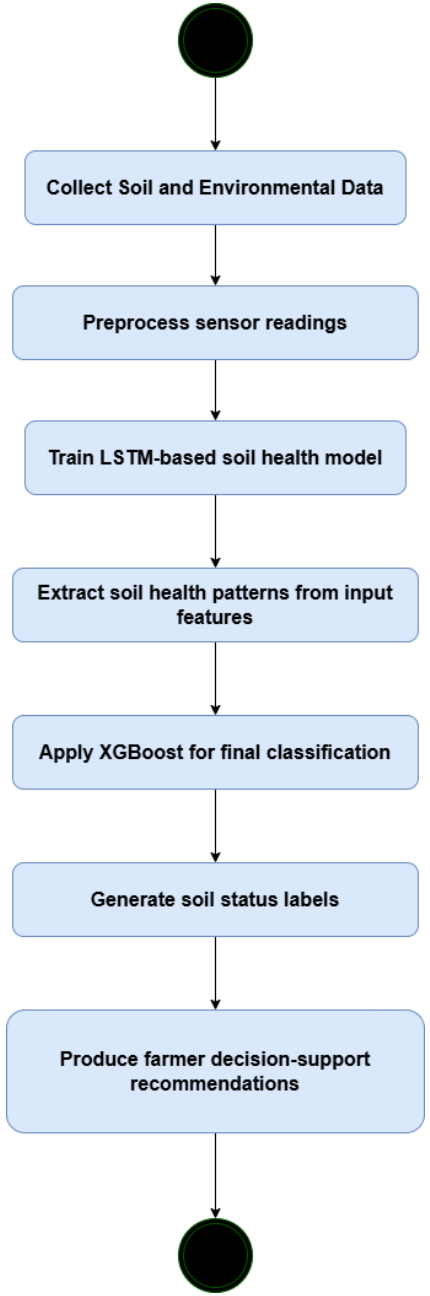


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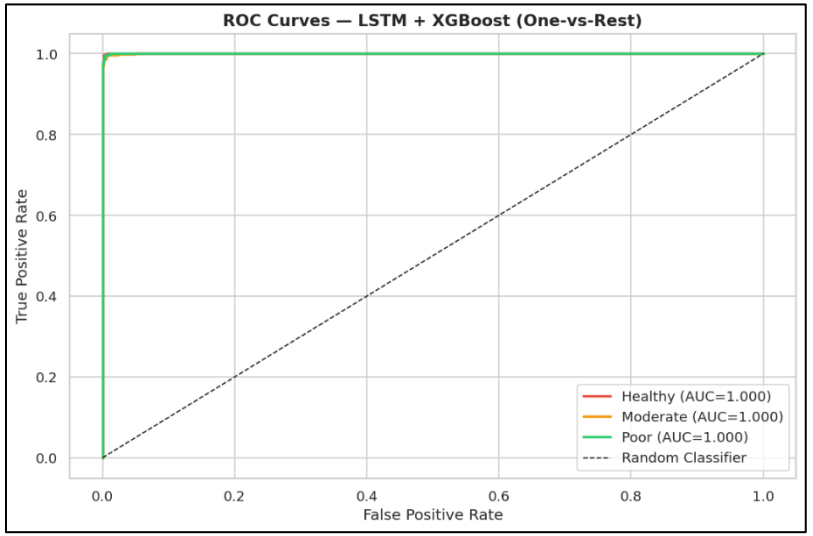
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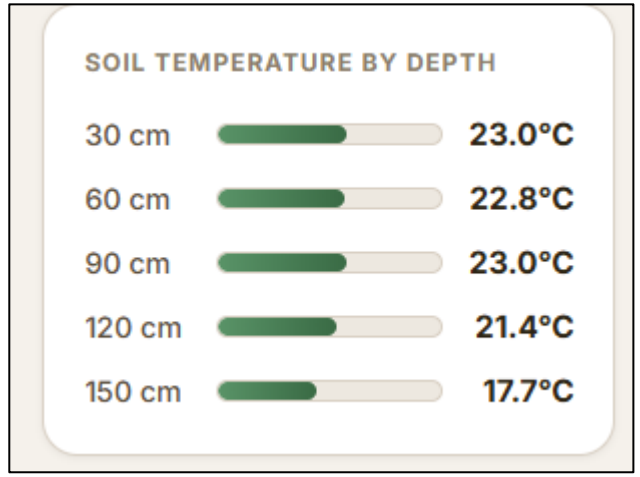
Methodology

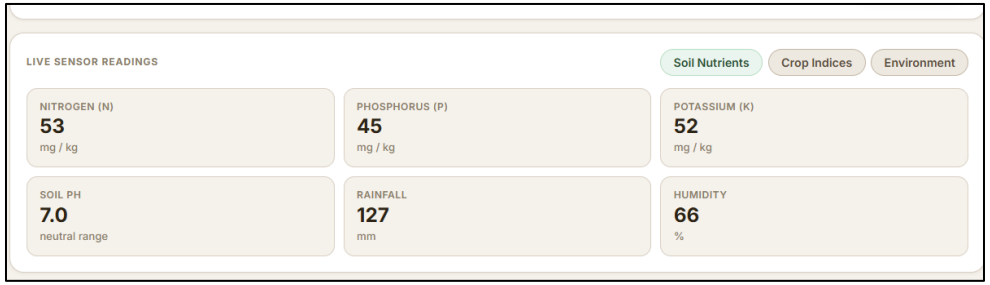


Completion of the project

- Soil health detection model was completed using NPK-based agricultural sensor data.20,000 records were processed and cleaned into 17,356 usable records.
- Soil health was classified into Healthy, Moderate and Poor categories using N, P, K, temperature, humidity, pH and rainfall.
- LSTM model achieved 88.12% test accuracy, while the LSTM + XGBoost hybrid model achieved 89.46% test accuracy.







Field Testing- Field Visit









HARTI-Guided Field Observation Visit

- The first field visit was conducted at a chilli plantation in Malagala, Padukka with guidance from Hector Kobbekaduwa Agrarian Research and Training Institute.
- The visit helped the team understand real chilli field conditions, including crop-row spacing, plant height, irrigation practices and common disease/stress symptoms.
- Field observations showed practical challenges such as uneven ground, narrow crop rows, sunlight variation, soil moisture differences and manual disease identification limitations.
- These findings were used to refine the rover movement plan, image capture angles, soil sensor placement and field data collection workflow.

Field Testing- Field Visit 2

HARTI-Guided Field Observation Visit

- The second field visit was conducted at a chilli plantation in the Hanwella–Avissawella area under HARTI guidance.
- This visit allowed the team to compare field conditions across two plantations and validate the need for zone-based crop monitoring.
- The team observed variations in soil condition, plant growth, irrigation coverage and crop health between different sections of the plantation.
- The visit confirmed the importance of integrating disease detection, soil health prediction, irrigation monitoring and field-zone mapping into one rover-based system.







Demonstration Video

What's next?

AGROROVER PROTOTYPE COMPARISON

P1 VS P2

PROTOTYPE 1		PROTOTYPE 2
Has some capture limitations due to the height of varieties.	CAMERA CAPTURE	Going to implement an antenna type camera which can adjust height on varieties.
Have difficulties to measure soil conditions on planted bags and maintain irrigation.	SOIL & IRRIGATION MEASUREMENT	Going to implement soil sensor as a stick to support both grounded and bagged plants.
Fixed camera height – not adjustable.	CAMERA HEIGHT ADJUSTABILITY	Antenna type camera allows height adjustment according to crop varieties.
Limited soil data collection – mainly for grounded soil.	SOIL SENSING CAPABILITY	Stick type soil sensor works for both grounded soil and planted bags.
Irrigation monitoring less effective on bagged plants.	IRRIGATION MAINTENANCE	Improved irrigation support for both grounded and bagged plants.
Lower versatility across different planting setups.	OVERALL VERSATILITY	More adaptable to various crop heights and planting environments.





Commercialization Plan

agroVER

Smart Rover. Smarter Farming.

COMMERCIALIZATION PLAN

agroVER delivers an intelligent agriculture monitoring solution that combines an autonomous field rover with a powerful **web dashboard** to help farmers and agribusinesses make smarter, data-driven decisions.



SOLUTION OVERVIEW

AGROVER ROVER



- ✓ Adjustable height camera (antenna type)
- ✓ Soil sensor stick for grounded and bagged plants
- ✓ Irrigation monitoring & control
- ✓ Real-time data & AI insights

AGROVER WEB DASHBOARD



- ✓ Live field monitoring
- ✓ Soil data & crop health analytics
- ✓ Irrigation control & scheduling
- ✓ Alerts & notifications
- ✓ Data history & reports
- ✓ Multi-user access

KEY BENEFITS

- Better Crop Monitoring
- Accurate Soil Measurement
- Efficient Irrigation Management
- Higher Yield & Profitability
- Time & Labor Saving
- Data-Driven Decisions

PRICING PLAN (SUBSCRIPTION)

BASIC PLAN
For small scale farmers

2,800 LKR / month

Includes:
- ✓ Web dashboard access (1 user)
- ✓ Live monitoring
- ✓ Basic analytics
- ✓ Irrigation alerts
- ✓ Data history (7 days)
- ✓ Email support

PREMIUM PLAN
For advanced farmers & plantations

30,000 LKR / month

Includes:
- ✓ Web dashboard access (Multi-user)
- ✓ All Basic Plan Features
- ✓ AI crop health analysis
- ✓ Advanced analytics & reports
- ✓ Irrigation control & scheduling
- ✓ Unlimited data history
- ✓ Priority support & training

ⓘ Note: Rover hardware is sold separately. Plans above are for web dashboard subscription & services.

ROVER HARDWARE PACKAGE

AGROVER ROVER UNIT — **LKR 450,000**

- ✓ Fully assembled agroVER rover
- ✓ Adjustable height camera (antenna type)
- ✓ Soil sensor stick
- ✓ Onboard controller & connectivity
- ✓ Irrigation monitoring module
- ✓ 1 Year warranty

*Shipping & installation charges may apply.



BUSINESS MODEL

- Hardware Sales (One-time)
- Subscription Plans (Recurring Monthly)
- After-Sales Support & Maintenance
- Future Add-ons & Upgrades

TARGET CUSTOMERS

- Small Scale Farmers
- Plantations & Large Farms
- Agricultural Companies
- Research & Institutions
- Government Programs

GO-TO-MARKET STRATEGY

- Direct sales & demos in farming communities
- Partnerships with agri cooperatives & dealers
- Online marketing & social media
- Training & onboarding for farmers
- Excellent after-sales service

OUR VISION

To revolutionize farming through smart robotics and data-driven insights, making agriculture more productive, sustainable and profitable for every farmer.



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